

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

API 306 - R

Combining Step Functions and EventBridge: Use cases and best practices

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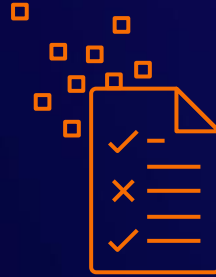
Principal Solutions Architect, Serverless
AWS



Traditional ways of workflow coordination



Coupling
applications using
synchronous APIs

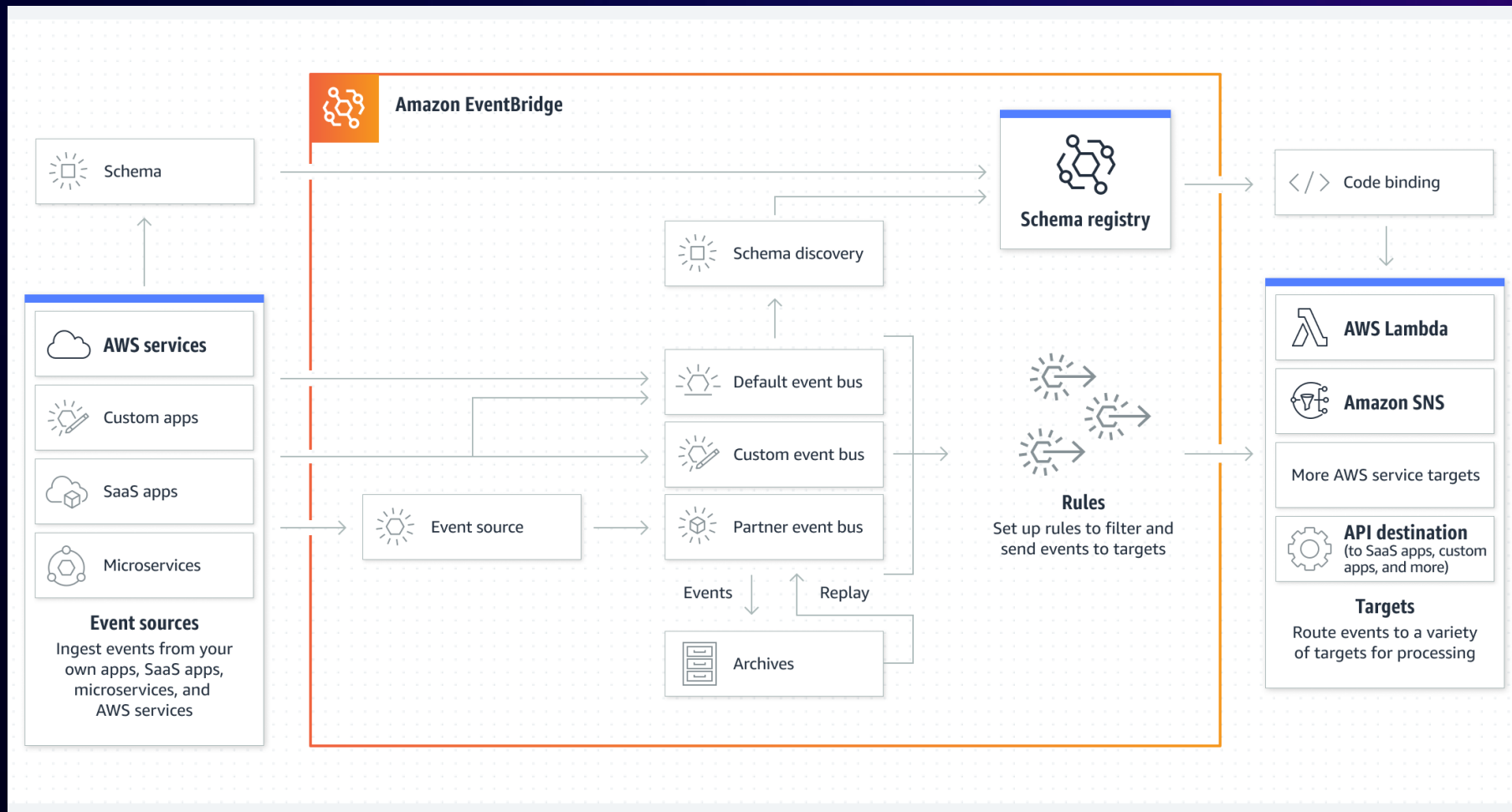


Coordination via
self-managed message
queues/topics

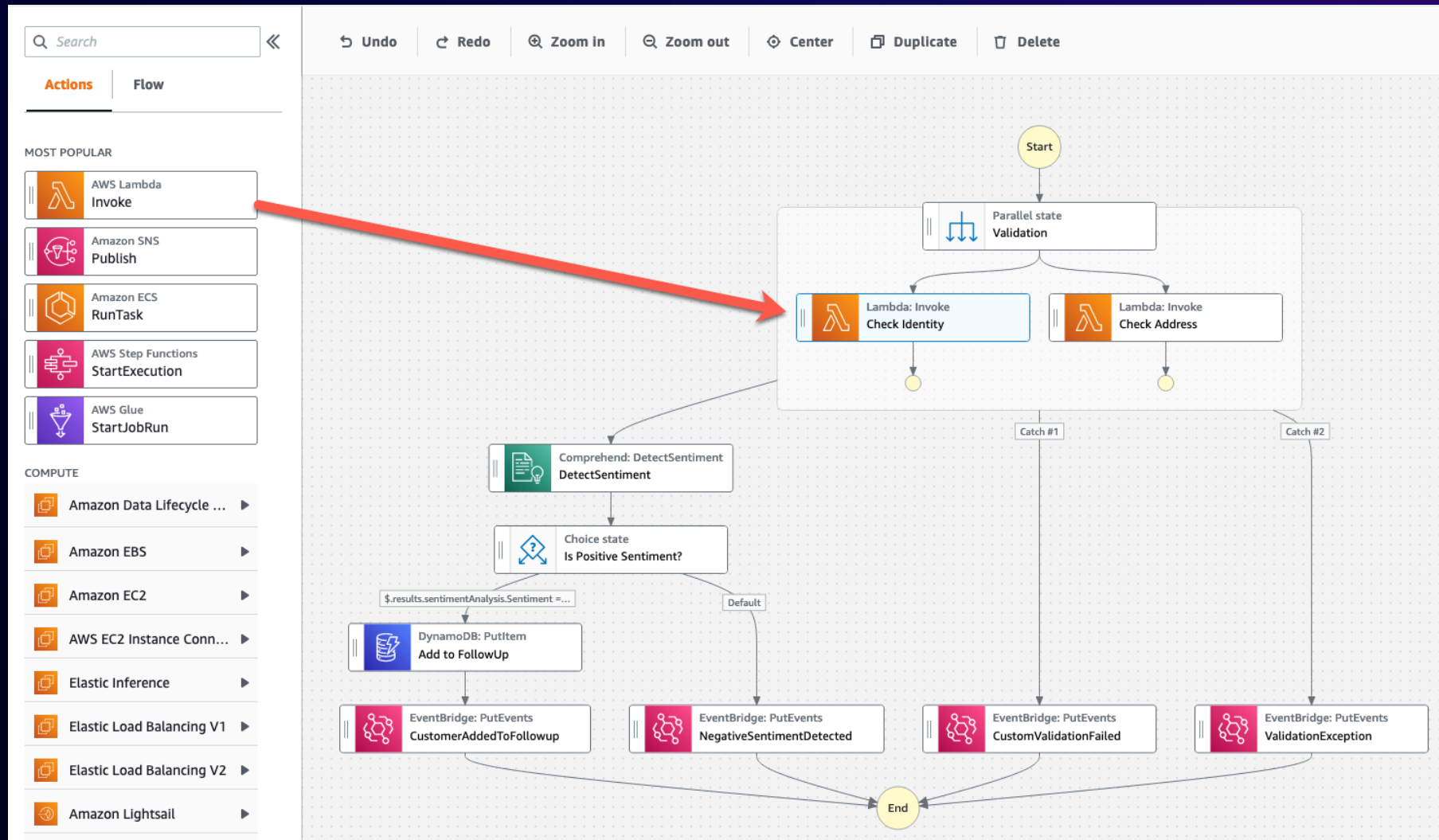


DIY orchestration in a
single monolithic
application

Amazon EventBridge

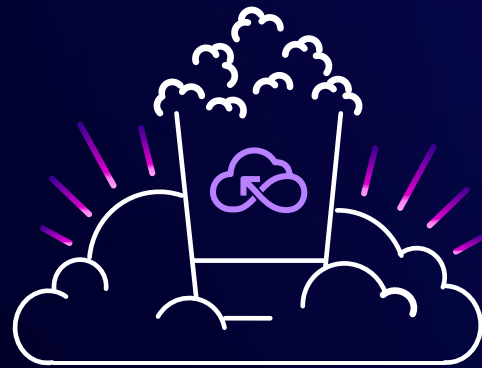


AWS Step Functions



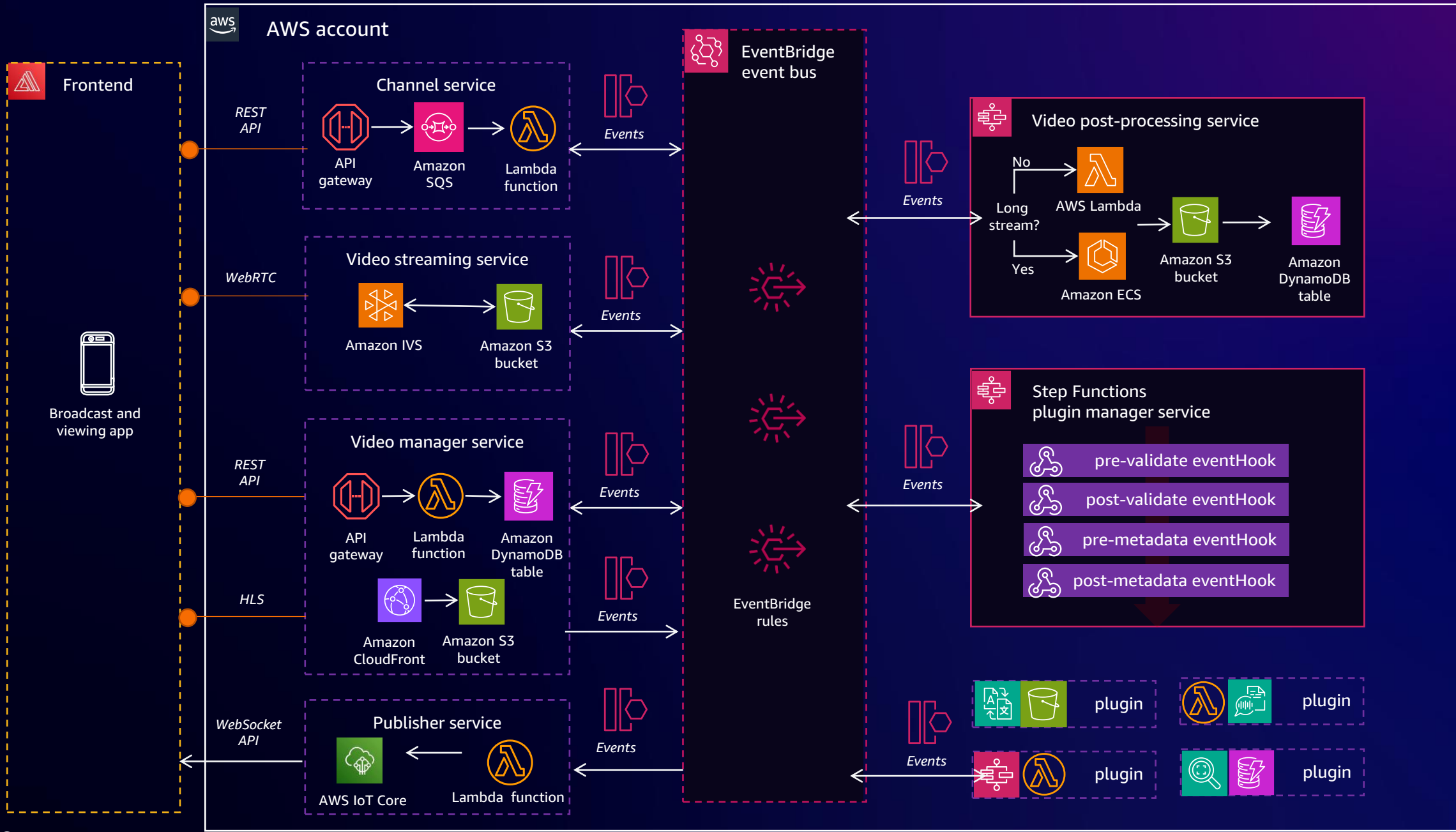
Use cases we will focus on today

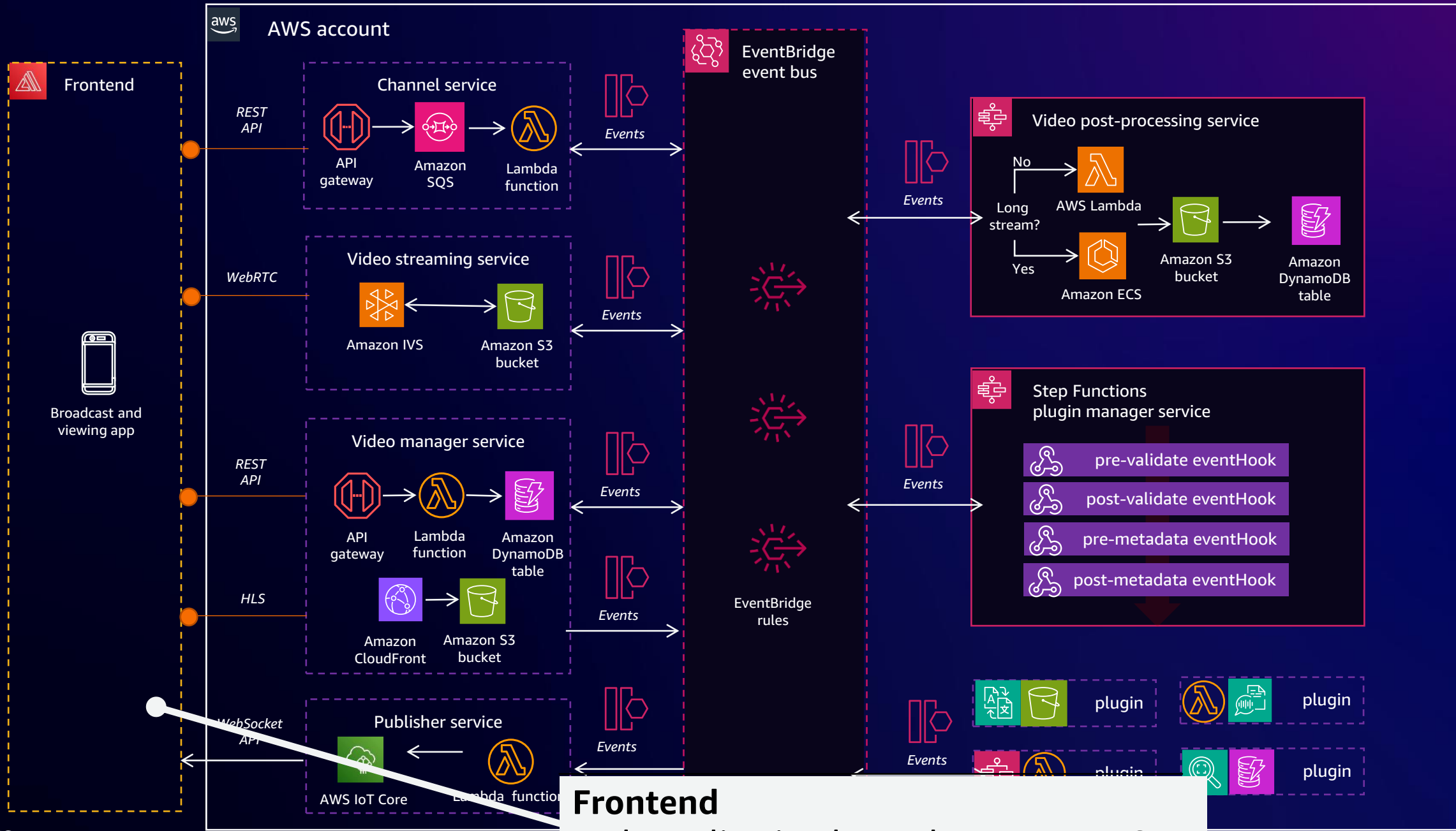
- Claims documents processing and its evolution
- Amazon S3 object processing at scale
- Streaming and on-demand video playback



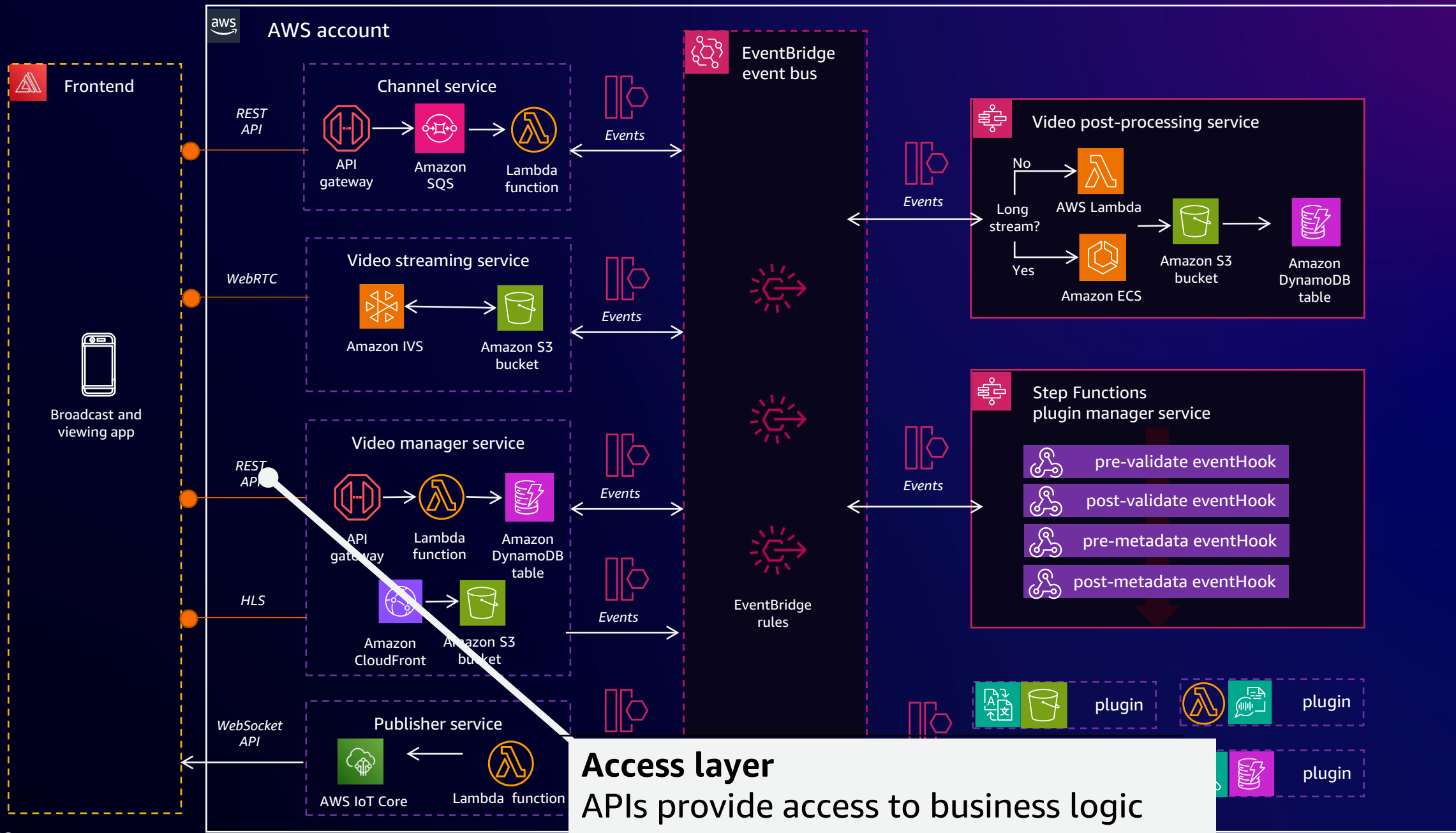
ServerlessVideo

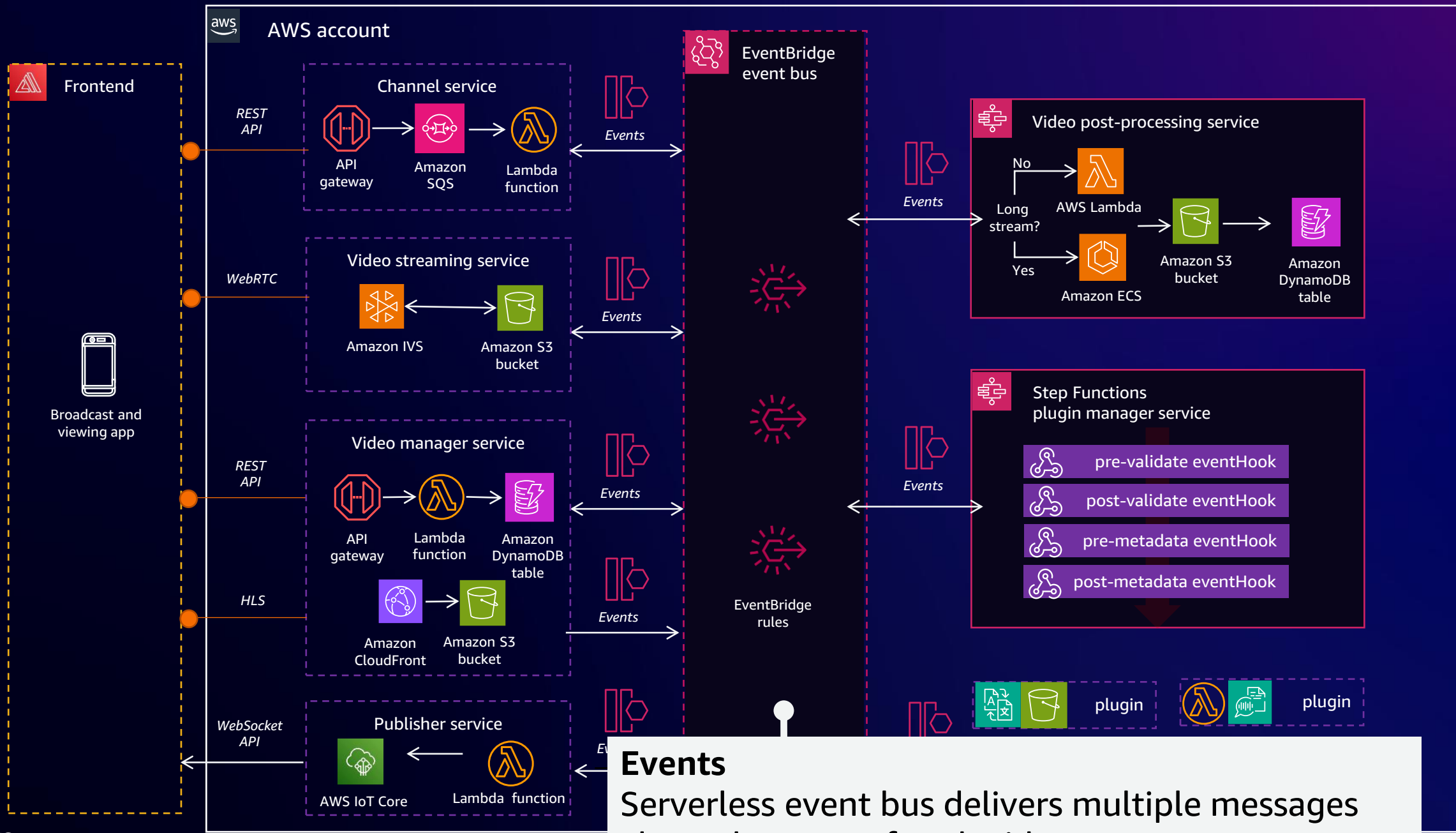






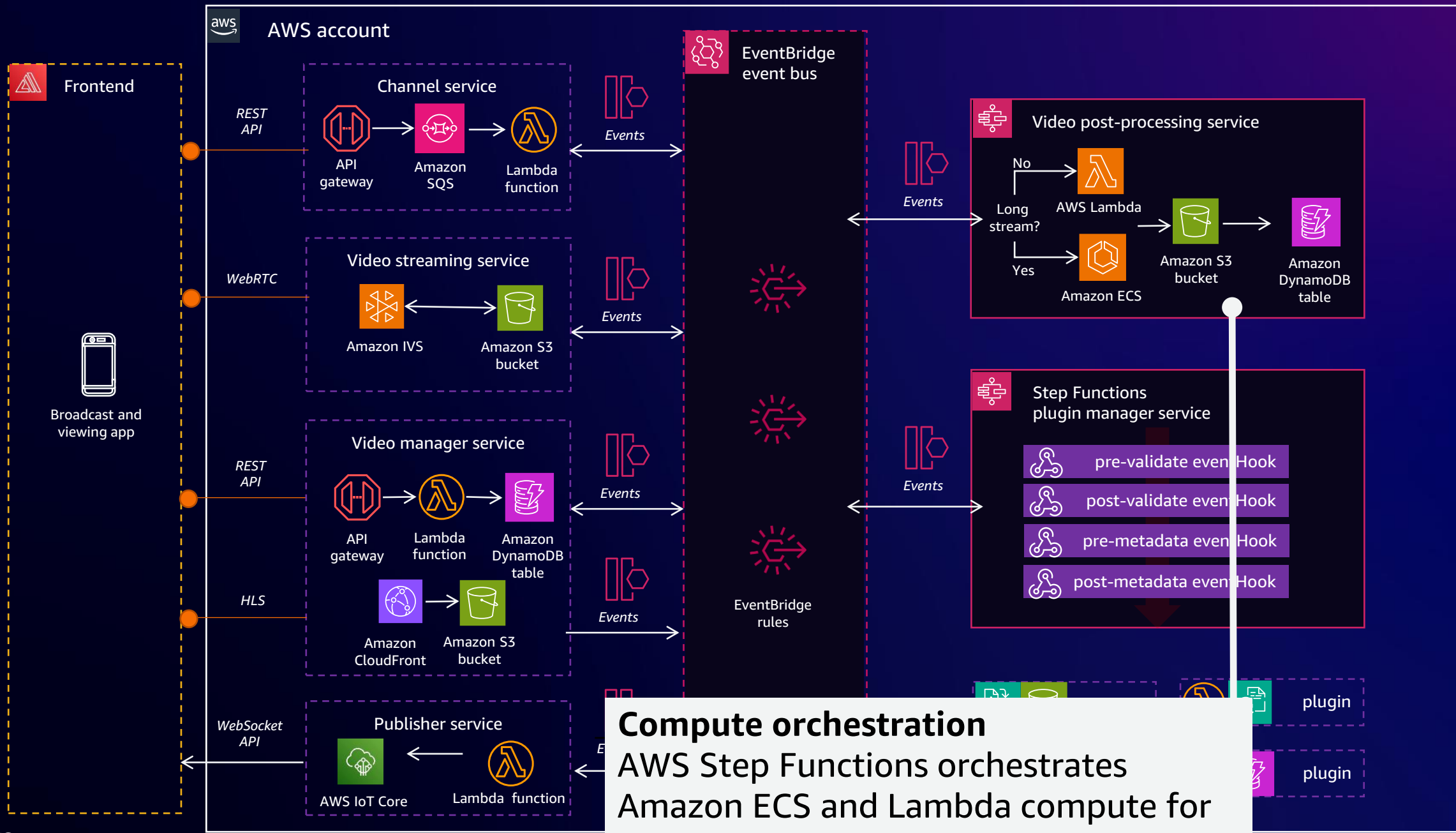
Frontend
Web application hosted on Amazon S3

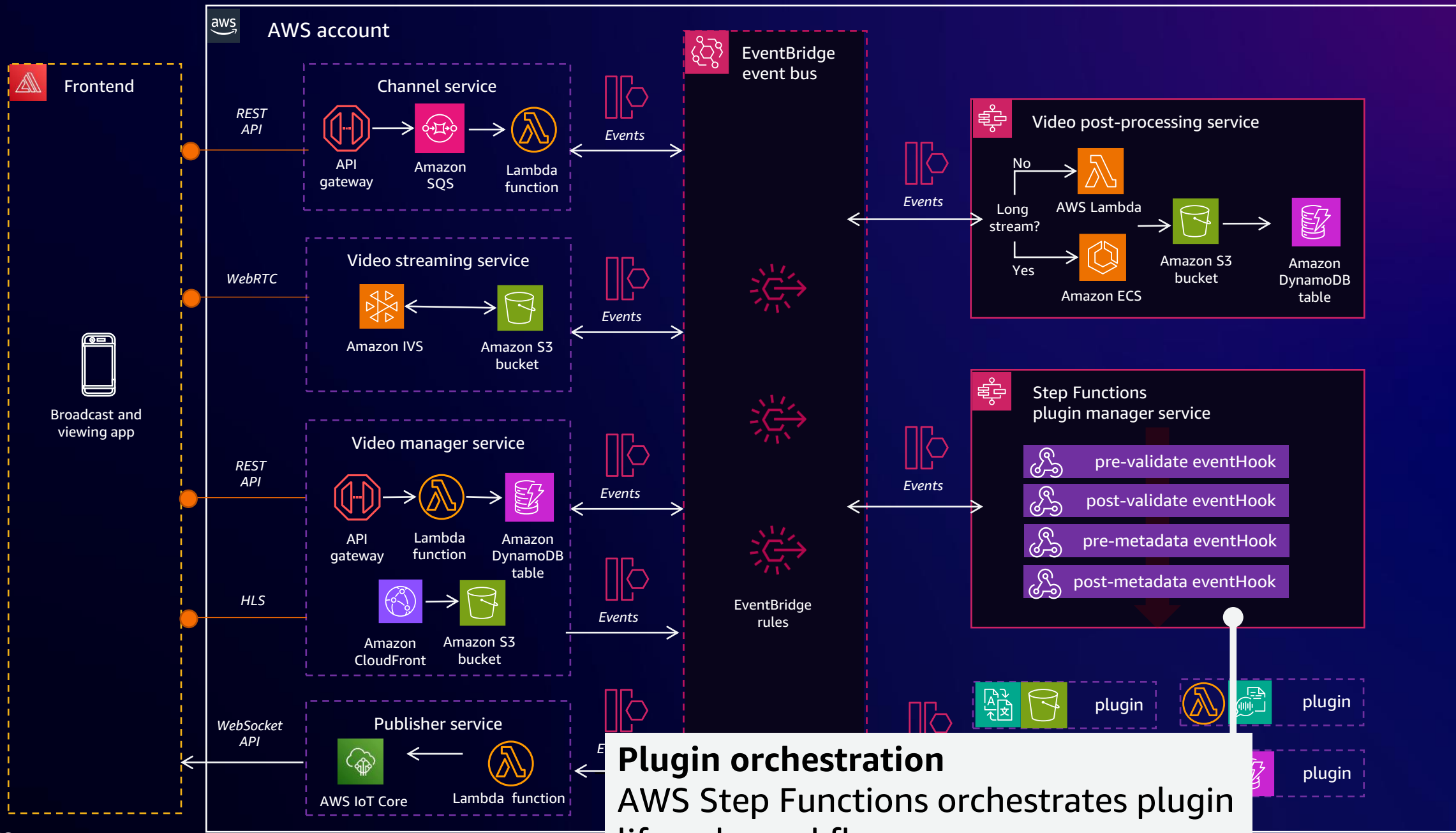


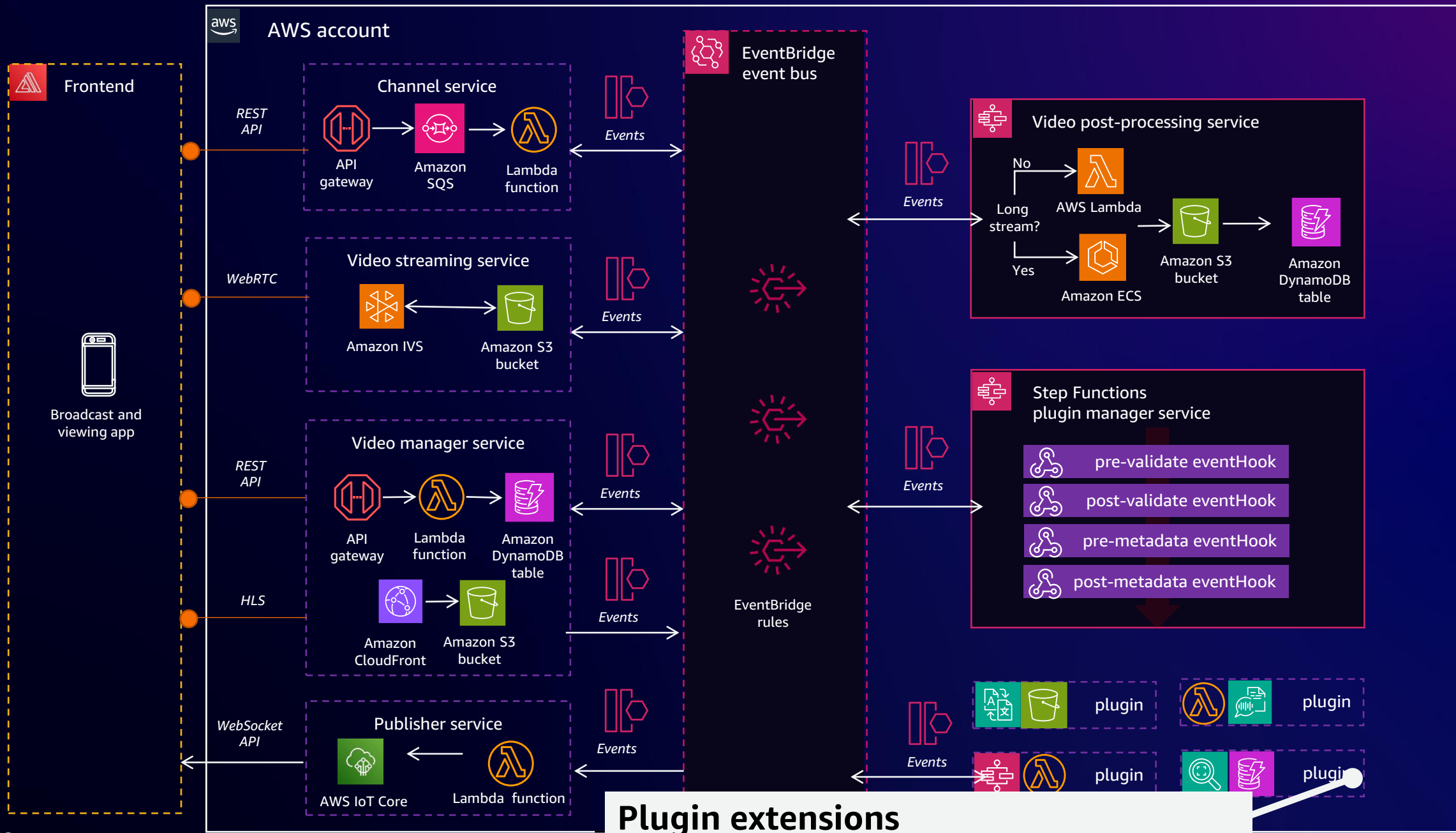


Events

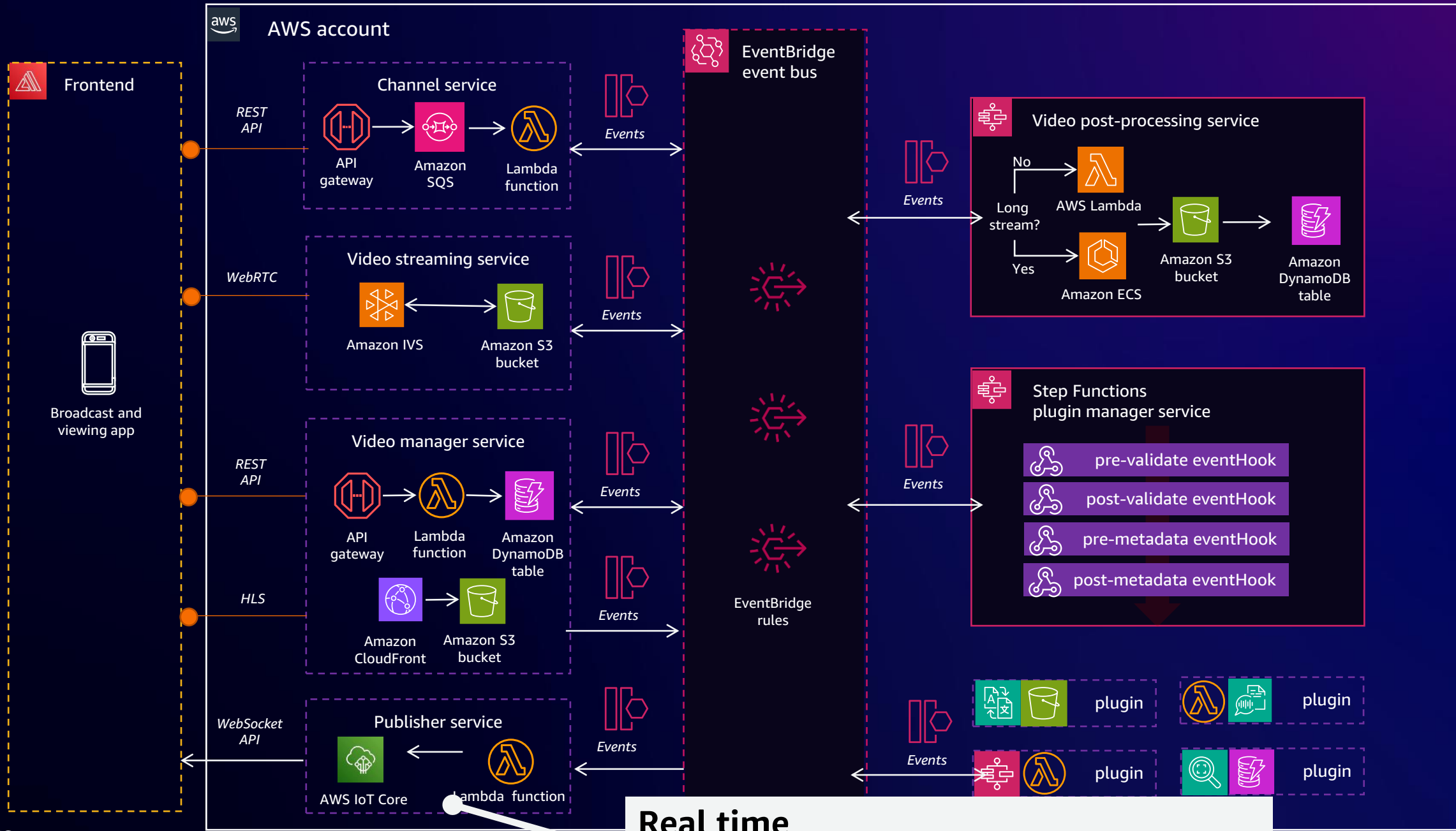
Serverless event bus delivers multiple messages about the state of each video stream



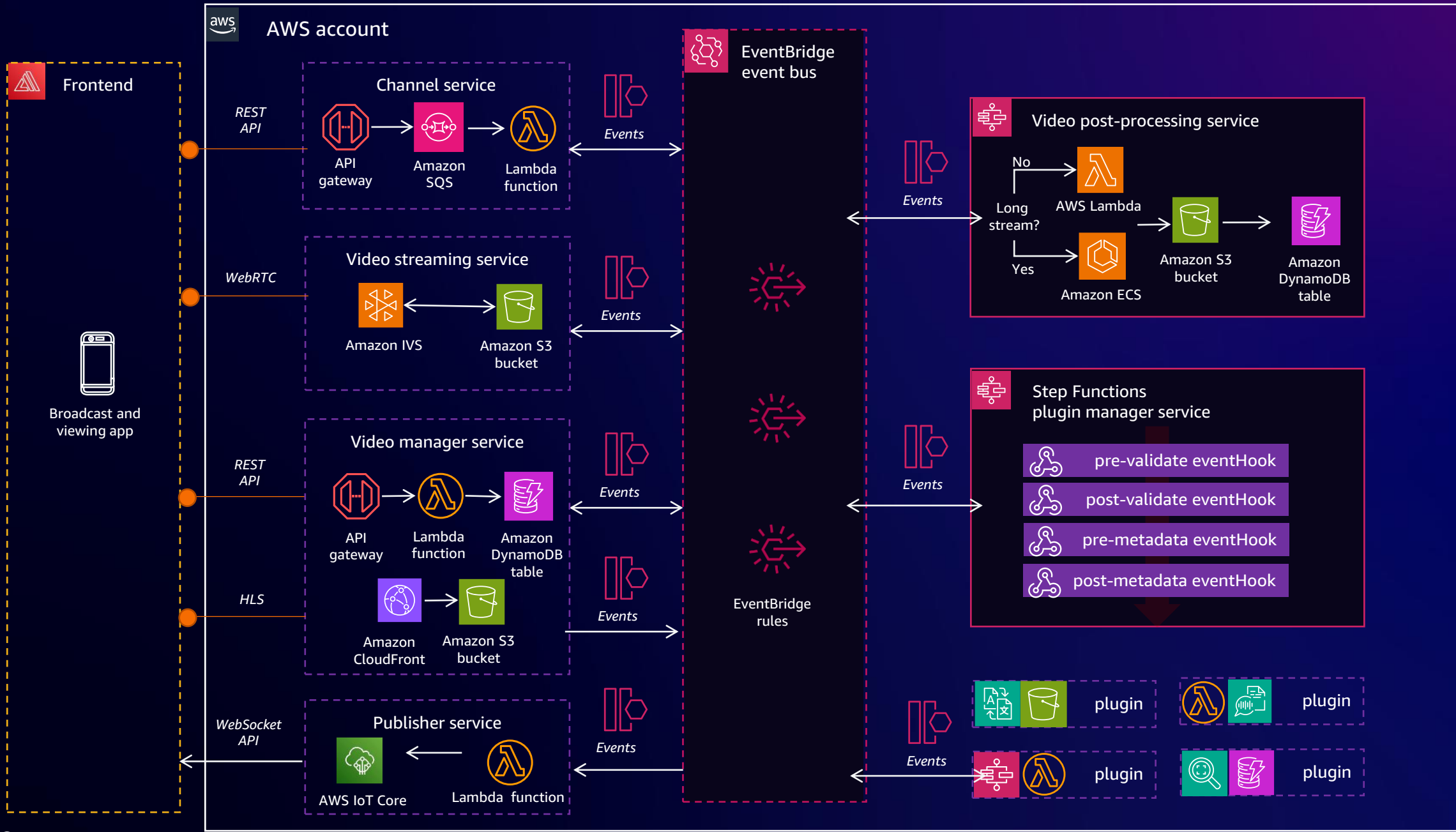




Plugin extensions
Plugins provide additional functionality



Real time
IoT topic updates web frontends

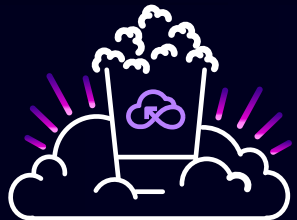


Extensible with plugin architecture

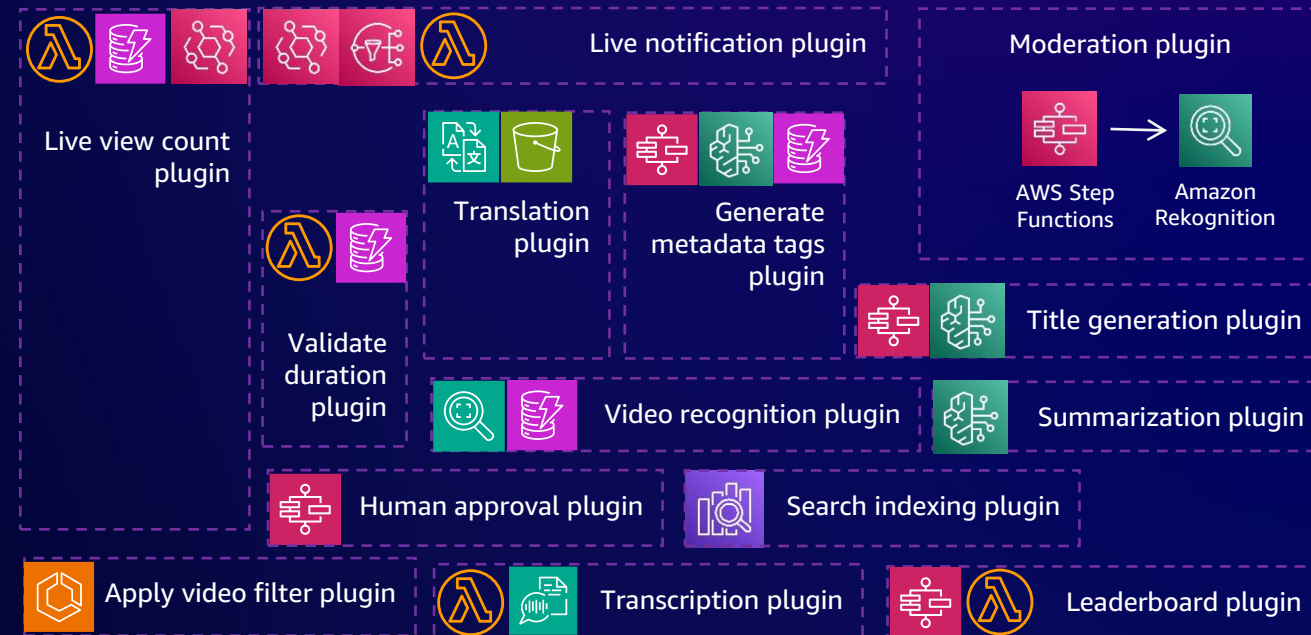
The core functionality is extended by multiple plugins

The plugins add capabilities for

- AI and ML workloads
- Human approval
- Notifications
- Translations
- Transcriptions
- Observability
- Searching
- Validation
- Reporting
- Video filters
- Leaderboards
- Gamification
- And more



ServerlessVideo



Machine learning and generative AI

A video title is generated from the text transcript using Amazon Bedrock

prompt = `generate a video title that does not contain any offensive words, for the following truncated transcript:"\${truncatedTranscript}..."`;

Title: "AWS Announces New Feature"

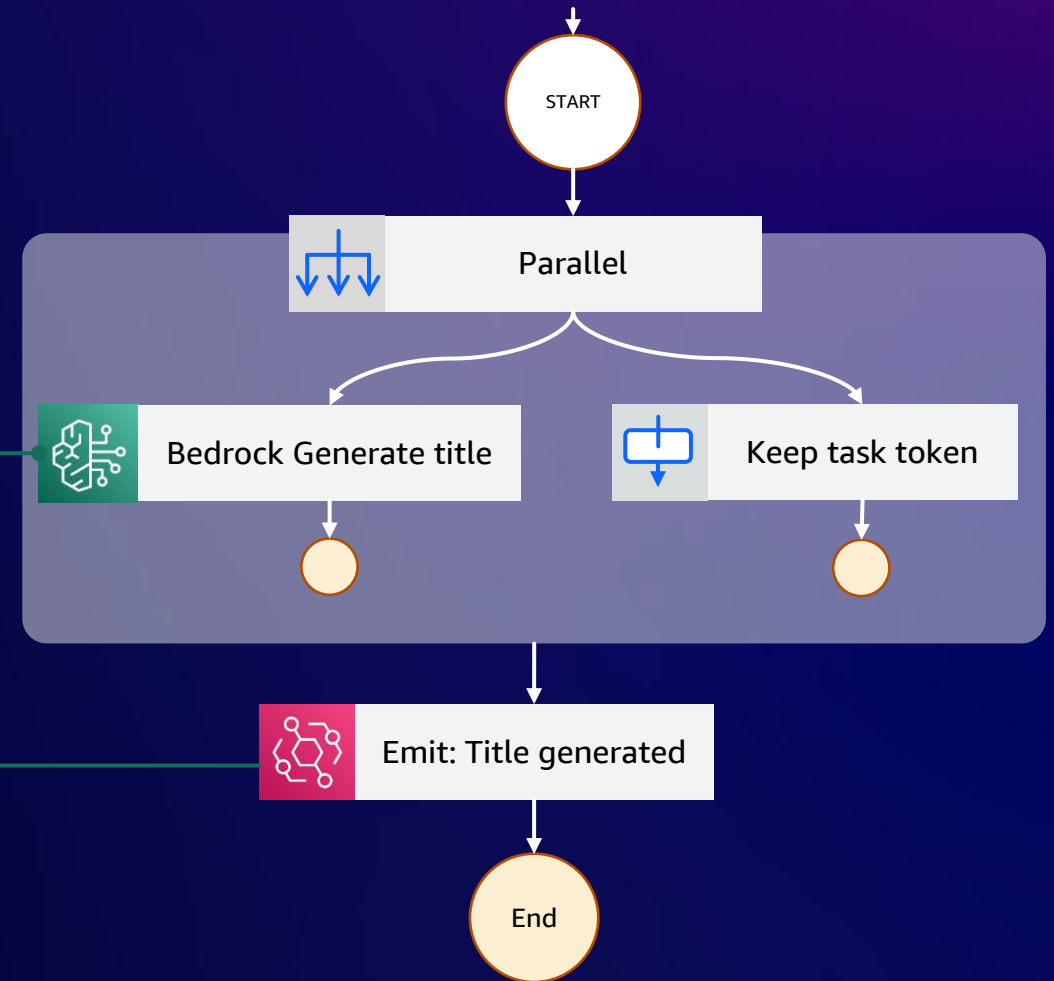


ServerlessVideo



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"Hello, everyone. Thanks for joining. I want to tell you all about the newest feature from AWS that was announced today ..."



Try it out! Go to: <https://s12d.com/video>

Multiple serverless compute substrates

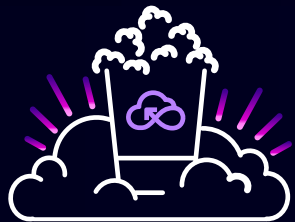
- Convert multiple TS files into a single .MP4
- Orchestrate multiple compute substrates with AWS Step Functions
- Amazon ECS for longer running tasks
- AWS Lambda for short duration
- Same code (minus the Lambda event handler)



```
- exports.handler = async (event) => {
```



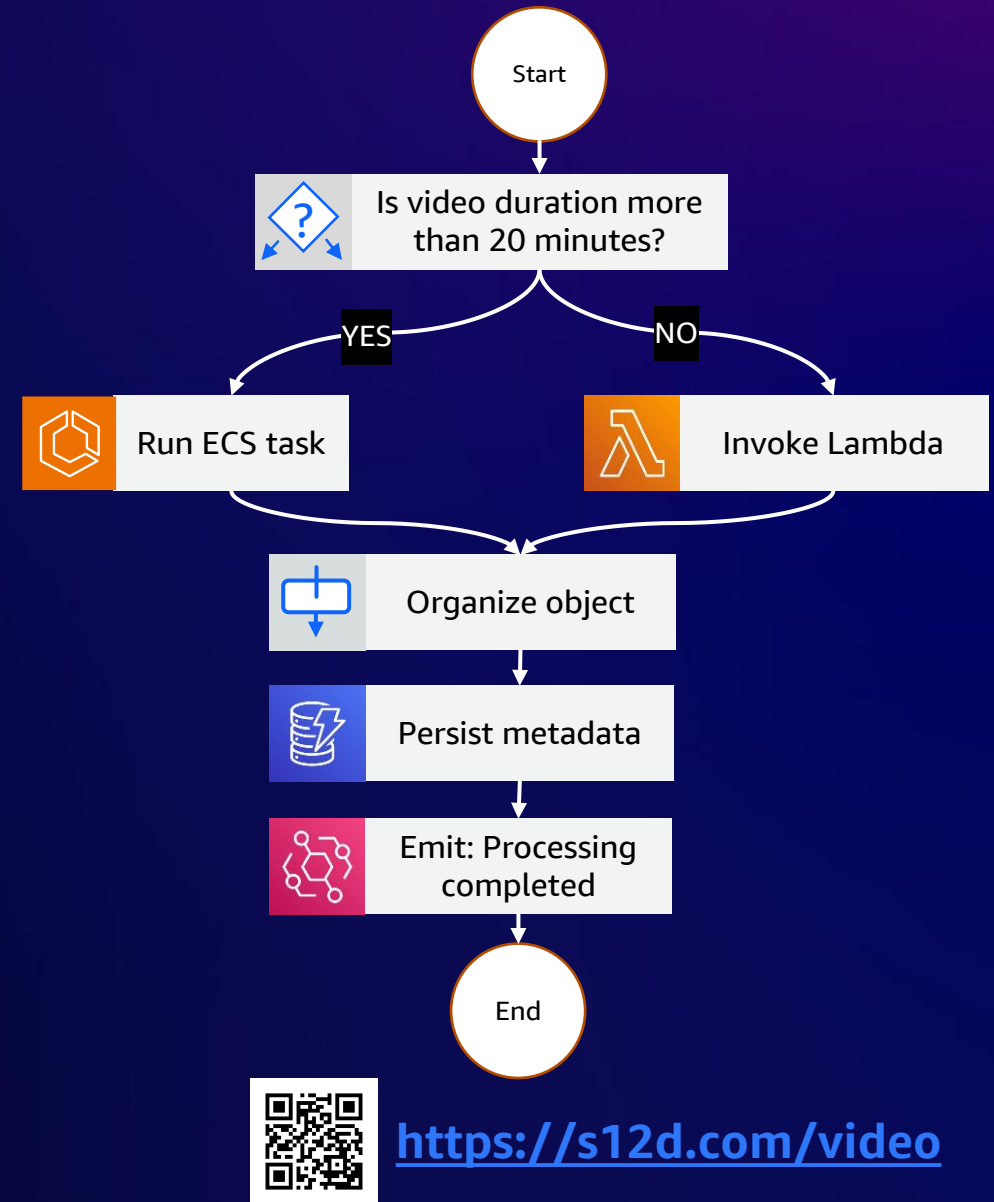
```
+ async function main() {
```



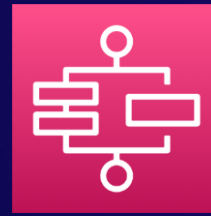
ServerlessVideo



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Takeaways



AWS Step Functions



Amazon EventBridge

- With serverless services like **Amazon EventBridge** and **AWS Step Functions**, you spend maximum time **implementing business value**
- Developers can maintain and **evolve** their solutions **without impacting their setup**
- These serverless services open up many possibilities with **hundreds of supported integrations** available in both services

Check out these other sessions

- **API206:** Bringing workloads together with event-driven architecture
- **API302:** Building next-generation applications with event-driven architecture
- **API303:** Navigating the journey to serverless event-driven architecture
- **API401:** Advanced serverless workflow patterns and best processes
- **SVS204:** Building Serverlesspresso: Creating event-driven architectures

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Best practices for **everyone**



Powertools for AWS Lambda

Python | TypeScript | Java | .NET

Batch processing

Config management

Observability

Secrets handling

REST/GraphQL API

Idempotency

Input/output validation

BYO middleware

Self-documented schemas

Feature flags

Caching

Streaming

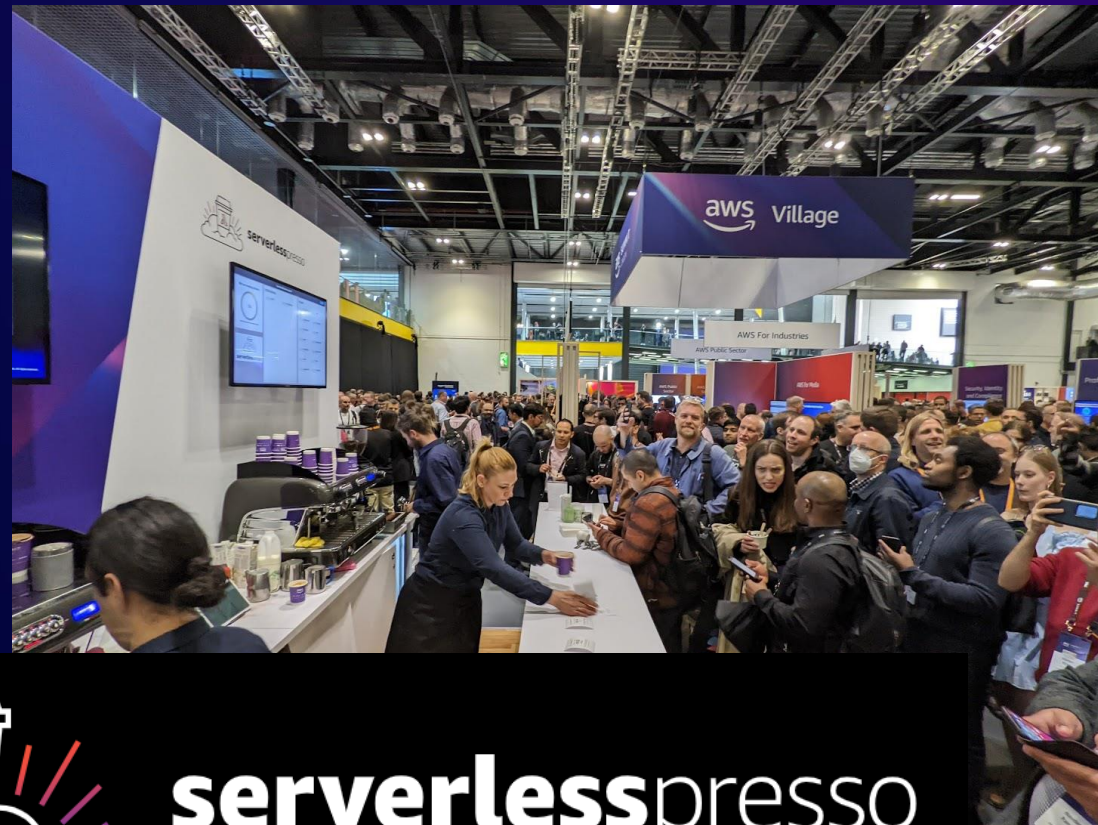
Data extraction

** Feature set may vary across languages*

Visit us in the Expo

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- Serverlesspresso
- AWS Modern Application and Open Source Zone



serverlesspresso

Resources



<https://s12d.com/api306>

Thank you!



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